

1) What is Java programming language? Discuss its key features and advantages.

↳ Java is a high-level, object-oriented programming language and computing platform. It was developed by Sun Microsystems in 1995. Java can be used in any devices given that they have a Java Virtual Machine (JVM) installed to run it. This programming language also has strong memory management and exception handling which allows for multithreading and concurrent execution, making it secure and reliable.

2) Explain the concept of platform independence in Java. How does Java achieve platform independence?

↳ Java being platform-independent means that it can run on any operating system as long as it has a JVM installed. A JVM, or a Java Virtual Machine, is Java's compiler, translating human-readable code into byte code. Each operating system has their own compatible version of a JVM.

3) Discuss the main components of a Java program and their roles.

↳ example code:

```
package com.example;

public class BSIT {

    public static void main(String[] args) {

        System.out.println("Hello World!");

        int num1 = 2;
        int num2 = 1;

        System.out.println("BSIT" + num1 + "-" + num2);

    }

}
```

Package → groups related classes and interfaces. ↳ helps organize code and prevent naming conflicts.

Class → defines variables and methods. ↳ serves as blueprint for creating objects in Java.

Method → a block of code that performs a specific task.

Statement → an instruction that performs an action.

variable → a named memory location used to store and manipulate data during program execution.

4) What are the differences between JDK, JRE, and JVM in Java?

↳ **JDK (Java Development Kit)** is a software development kit used to build Java applications.

↳ it contains the JRE, a set of developmental tools, a compiler, and a debugger.

- ↳ it's only used for development and does not need to run java programs.
- ↳ is platform-dependent (has different version for every operating system).

↳ **JRE (Java Runtime Environment)** provides an environment to run Java programs.

↳ it does not include development tools, contains only JVM and standard class libraries.

↳ provides all runtime requirements for java applications and does not support compilation or debugging.

↳ Intended for users who only need to execute applications.

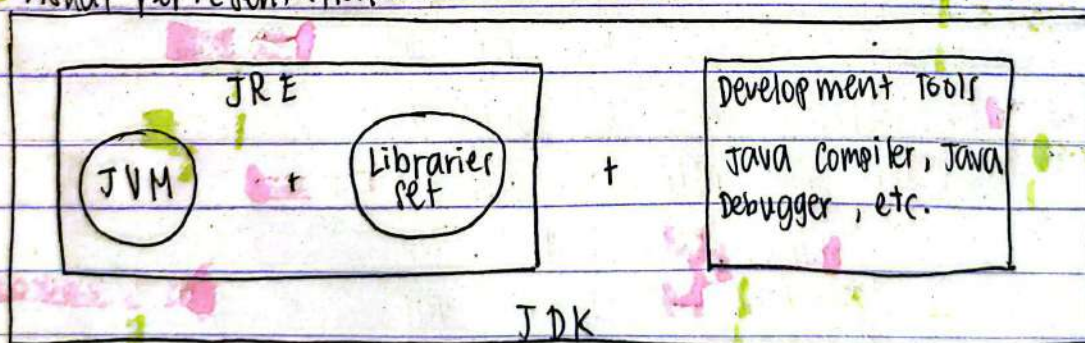
↳ **JVM (Java Virtual Machine)** is the core execution engine of Java.

↳ it is part of both JDK and JRE and is responsible for converting byte code into machine-specific instructions.

↳ performs memory management and garbage collection.

↳ its implementation is platform-dependent.

↳ visual representation:



5) Explain the difference between interpreted and compiled languages. Is Java an interpreted or compiled language?

↳ Java is both an interpreted and compiled language. The source code is first compiled into byte code. Then, the JVM interprets the bytecode and translates it into machine code which the underlying hardware can execute. The JVM makes it possible for Java to be platform-independent.